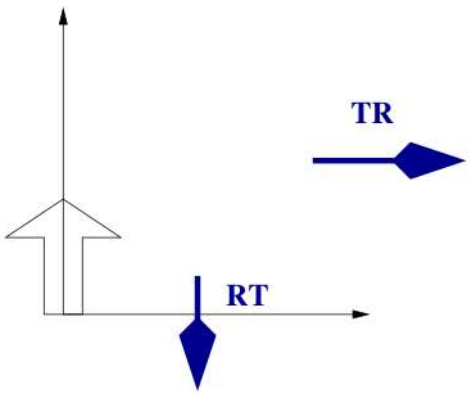
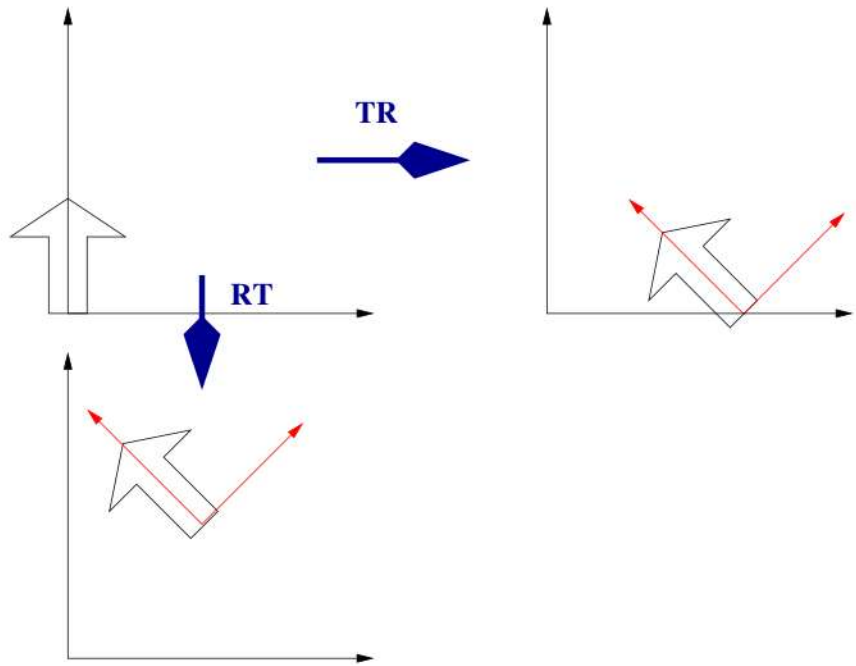
Order of Transformation is Important



Order of Transformation is Important

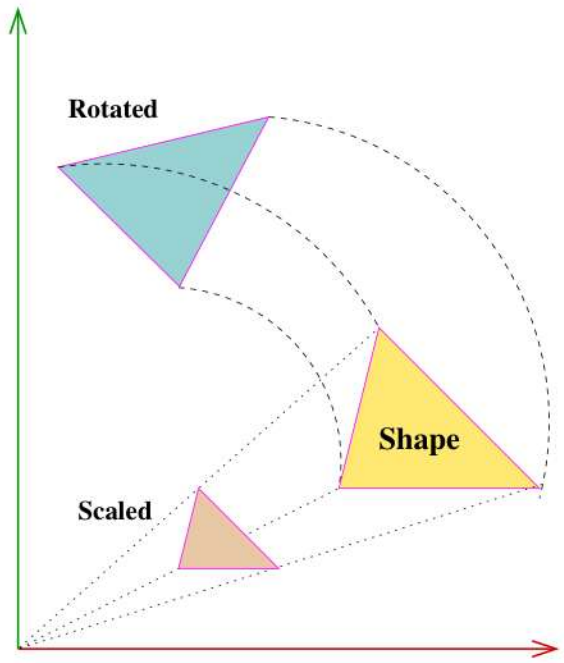
$TR \neq RT$



TR keeps it on X axis to $(5, 0)$. **RT** takes it to $(\frac{5}{\sqrt{2}}, \frac{5}{\sqrt{2}})$.

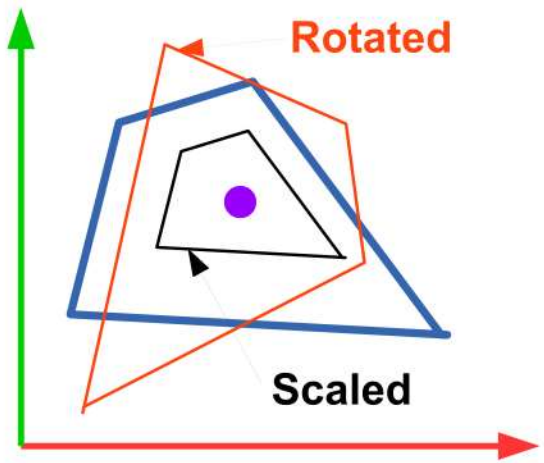
Objects Away from Origin

- Object "translates" when rotated or scaled!!
- Default: Perform these **about the origin**
- How do we transform points "in place"?
- Rotate or scale about the centroid of the object. Or about an arbitrary point
- How?



Transformation About a Point

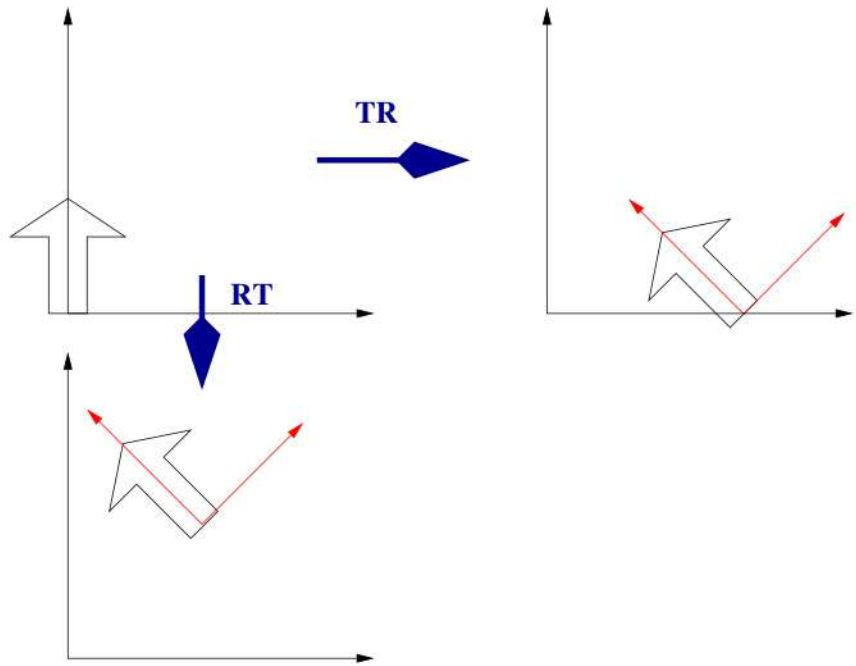
- Rotating about point P
  – Align P with origin
  – Rotate/scale about origin
  – Translate back
- Rotation:
  $\mathbf{R_C}(\theta) = \mathbf{T}(\mathbf{C})\ \mathbf{R}\ \mathbf{T}(-\mathbf{C})$
- Scaling:
  $\mathbf{S_C}() = \mathbf{T}(\mathbf{C})\ \mathbf{S}()\ \mathbf{T}(-\mathbf{C})$
- A transformation $\mathbf{M}$:
  $\mathbf{M_C} = \mathbf{T}(\mathbf{C})\ \mathbf{M}\ \mathbf{T}(-\mathbf{C})$



Operation on Points

- **T(5,0) R($\pi/4$)**: Impact on a point:
  - R($\pi/4$): **(Point stays at $(0,0)$)**
  - T$(5,0)$: **(Point goes to $(5,0)$)**
- **R($\pi/4$) T(5,0)**: Impact on the point:
  - T$(5,0)$: **(Point moves to $(5,0)$)**
  - R($\pi/4$). **(Point rotates about origin)**
- All points on the shape undergo the same motions and get new coordinates

Order of Transformation is Important



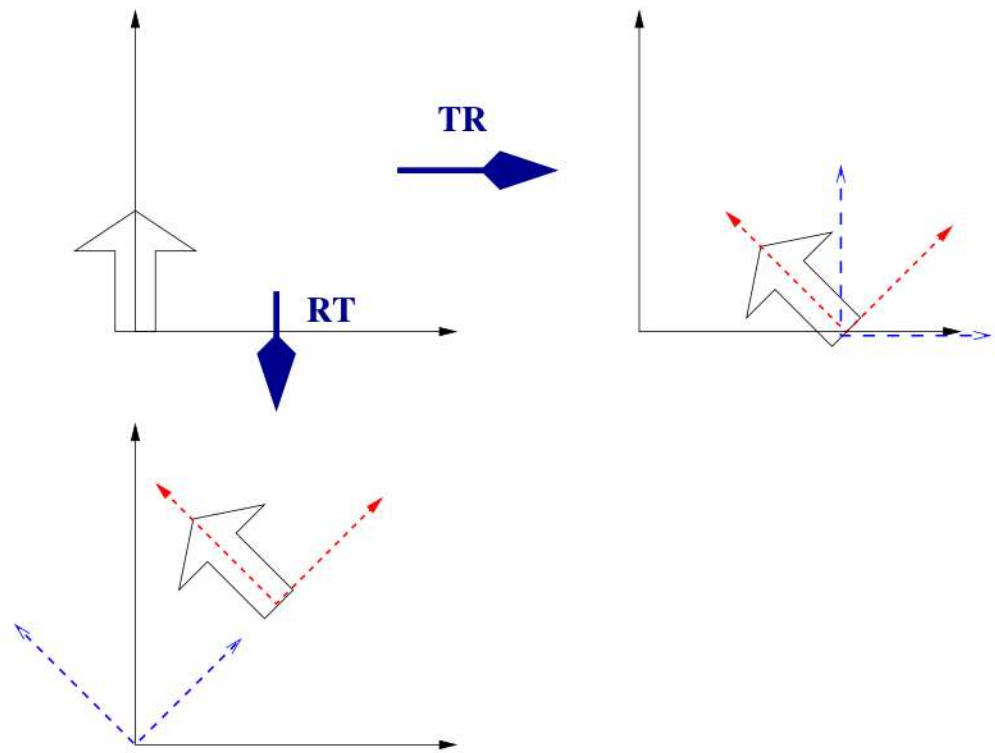
TR keeps it on X axis to $(5, 0)$. **RT** takes it to $(\frac{5}{\sqrt{2}}, \frac{5}{\sqrt{2}})$.

Operation on Frames

- **T(5,0) R($\pi/4$)**: Impact on coordinate frame:
  - T$(5, 0)$: **(Origin shifted to $(5, 0)$)**
  - R$(\pi/4)$. **(Axes rotated at new origin)**
- **R($\pi/4$) T(5,0)**: Impact on coordinate frame:
  - R$(\pi/4)$: **(Axes rotate by 45 degrees))**
  - T$(5, 0)$. **(Point moves to $(5, 0)$ in new axes)**
- Frames move around, giving new coordinates to points on objects!!

Relating Coordinate Frames

- $T(5,0)$ and $R(\pi/4)$
- Start: Black axes
  Next: Blue axes
  Last: Red axes

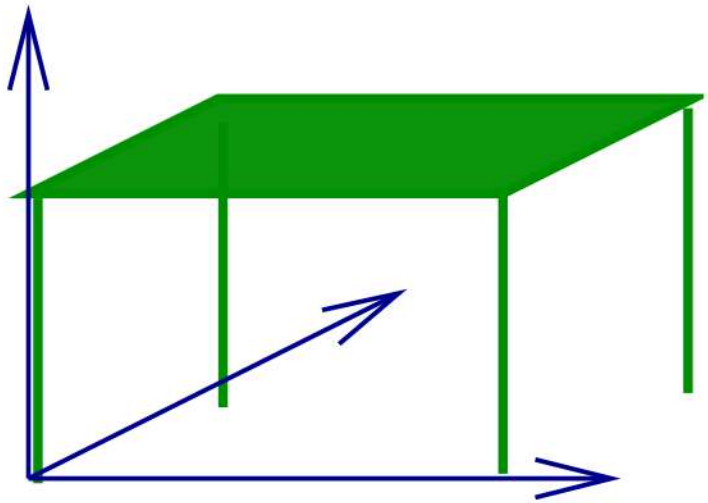


Coordinates v/s Frames

- Consider $P_4 \ = \mathbf{M_4 M_3 M_2 M_1} \ \ P_0$
- Point P_0 undergoes 4 operations and get coordinates P_4
- Imagine the point having coordinates P_1, P_2, P_3 after operations $\mathbf{M_1}, \mathbf{M_2}, \mathbf{M_3}$
- We can also visualize coordinate frames $\mathbf{\Pi}_4, \mathbf{\Pi}_3, \mathbf{\Pi}_2, \mathbf{\Pi}_1, \mathbf{\Pi}_0$ in which a point has coordinates P_4 to P_0 respectively
- Operation $\mathbf{M}_i$ represents a change in coordinates from $\mathbf{\Pi}_i$ to $\mathbf{\Pi}_{i-1}$, resulting in new labels for the point.

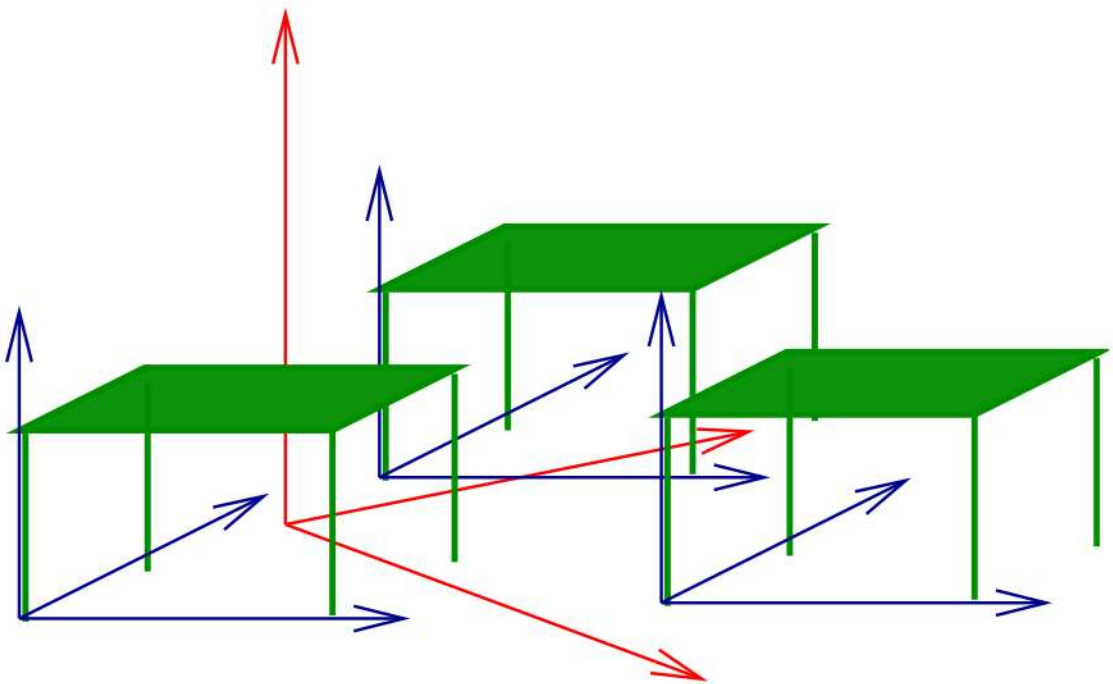
Aligning Multiple Coordinate Systems

A table defined in its own coordinate system.



Aligning Multiple Coordinate Systems

Place many tables in a **common world coord system**!



Aligning Multiple Coordinate Systems

- Model IITJ Campus as a whole. Campus is our “world”.
- Global coordinate frame $\mathbf{\Pi}_G$ for the campus: at the Gate
- Buildings: LHB, CC, Library,.., CSE. Each has a natural coordinate frame. $\mathbf{\Pi}_H$ is of CSE building.
- CSE: CSE has several rooms: 101, 102, 104, 201 etc., with own coordinate frames. $\mathbf{\Pi}_C$ is of 104 (our class)
- 104 has 55 desks, with coord frames $\mathbf{\Pi}_{Di}$ for desk i
- Desks are identical in geometry; the coord frame Π_{Di} places each in its location.

Aligning Multiple Coordinate Systems

- Consider a corner point P of desk 37, with coordinates $(200, 30, 100)$ in $\mathbf{\Pi}_{D37}$. That is: $P_{D37} = (200, 30, 100)$
- Since our world is the campus, we have to ultimately describe everything in the coordinate frame $\mathbf{\Pi}_G$

$$P_G = \mathbf{M_{GH}}\ \mathbf{M_{HC}}\ \mathbf{M_{CD37}}\ P_{D37}$$

- $\mathbf{M_{GH}}$ aligns $\mathbf{\Pi}_G$ to $\mathbf{\Pi}_H$. $\mathbf{M_{HC}}$ aligns $\mathbf{\Pi}_H$ to $\mathbf{\Pi}_C$. $\mathbf{M_{CD37}}$ aligns $\mathbf{\Pi}_C$ to $\mathbf{\Pi}_{D37}$
- $P_G = \mathbf{M_{GH}}\ \overset{P_H}{|}\ \mathbf{M_{HC}}\ \overset{P_C}{|}\ \mathbf{M_{CD37}}\ \overset{P_{D37}}{|}\ P$ (for any point P on Desk37)
- We can place a given desk in any **building, room, place**!

Walking on Stage

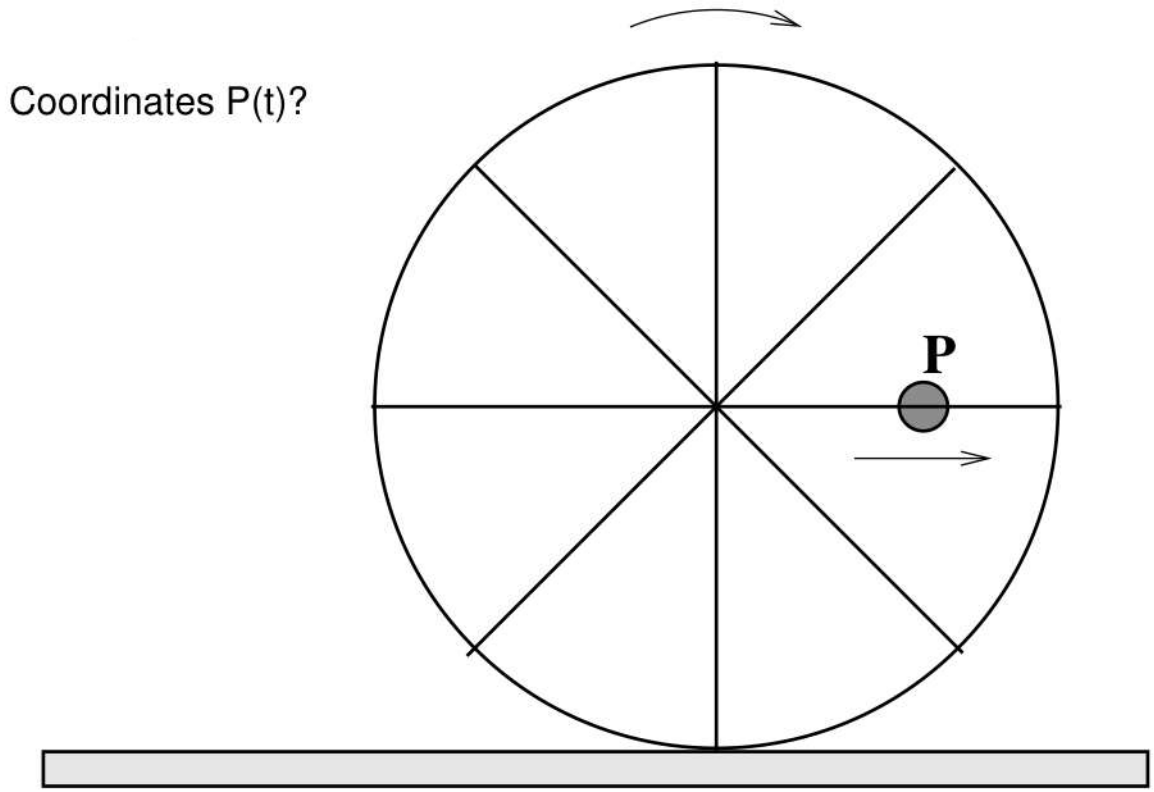
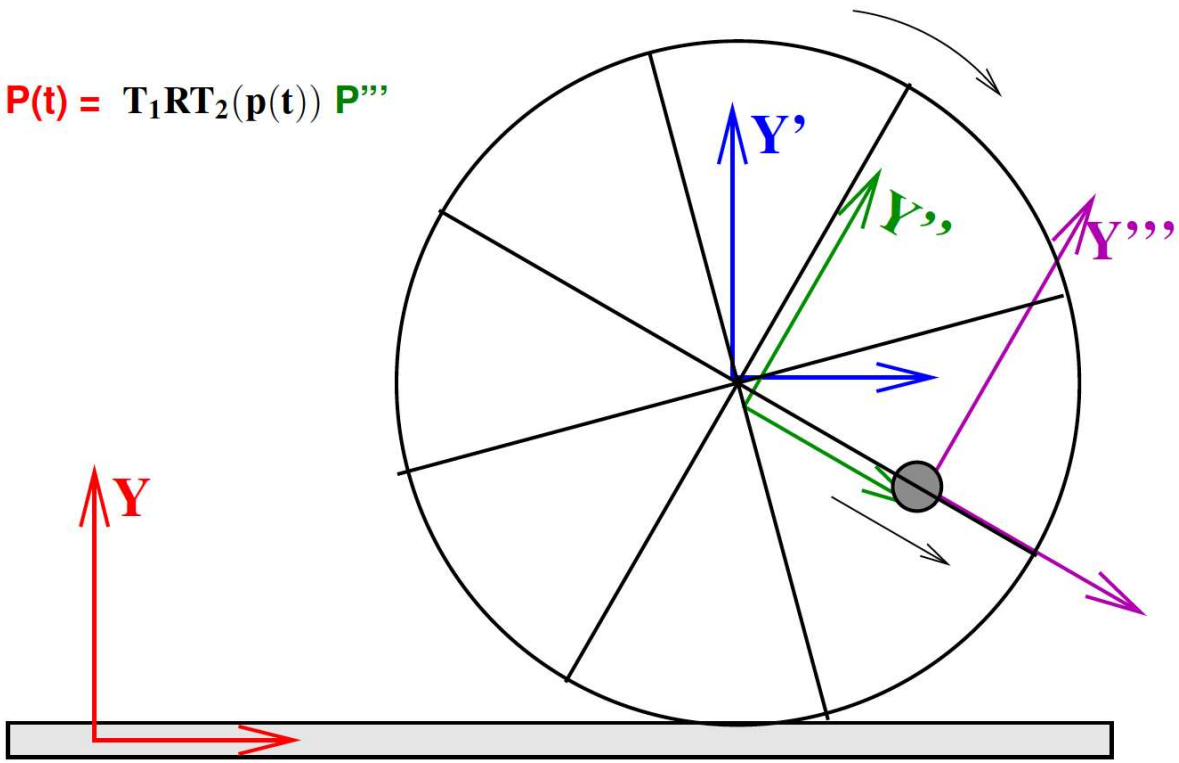
- Person walking horizontally on stage, with swinging arms
- How does the hand-tip move w.r.t each student? **How?**

Walking on Stage

- Person walking horizontally on stage, with swinging arms
- How does the hand-tip move w.r.t each student? **How?**
- Student knows own position in room's reference frame
- Start at a student's eye. (That provides the viewpoint!)
- Align to room's reference frame using $\mathbf{M_1}$. Different matrix for each student, but everyone same now....
- Walk: pure translation. $\mathbf{M_2}$ aligns to person coord frame
- Arm swing: Simple harmonic motion with angle $\theta(t)$

Simpler viewpoints in newer coord frames.

Aligning Multiple Coordinate Systems



Aligning Multiple Coordinate Systems